ADITYA PILLAI

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Education

Indian Institute of Information Technology Design and Manufacturing Kancheepuram — Chennai
CGPA: 9.36/10 *B.Tech (Hons.) in Computer Science and Engineering* — *2022 – 2026*

- **Relevant Coursework:** DSA, OOP, DBMS, Operating Systems, Computer Networks & Security, Algorithms, Cloud Computing, Artificial Intelligence & Machine Learning **Class X:** 95.2% **Class XII:** 94.8%

Technical Skills

- **Languages:** Java, Python, TypeScript, JavaScript, C, C++, SQL, PowerShell
- **Backend & Frontend:** Spring Boot, Spring Cloud, Spring AI, FastAPI, Flask, Node.js, Next.js, React, REST APIs, Microservices, OAuth
- **Cloud & DevOps:** AWS, Azure (Functions, SQL, DevOps, Automation), Docker, Terraform, Jenkins, CI/CD, Agile
- **AI / ML:** LLMs, RAG, LangChain, Qdrant, Hugging Face, PyTorch, TensorFlow, Scikit-learn, Computer Vision
- **Data & Testing:** MongoDB, Pandas, NumPy, Power BI, Jupyter, Unit Testing, Integration Testing

Experience

Accenture — India
Java FDE — *Jul 2026 – Present*

- Building **AI-first, cloud-native enterprise applications** on **Java, Spring Boot, and Spring Cloud** microservices, deployed on **AWS**, with AI capabilities embedded into core product workflows from the outset.
- Integrating **LLM-powered features using Spring AI** (RAG, tool calling) into backend services, and delivering responsive **Next.js / React** frontends over secure, well-tested REST APIs.
- **Tools:** *Java, Spring Boot, Spring Cloud, Spring AI, Next.js, React, AWS, Docker, CI/CD*

Quantrium — Chennai, India
Machine Learning Intern [Certificate] — *Jan 2026 – June 2026*

- Architected a **multi-tenant, production-grade chatbot platform** serving multiple enterprise clients, designing **scalable system architecture** with fault tolerance and low-latency query handling at scale.
- Built a **distributed RAG pipeline** with real-time semantic indexing and dynamic corpus updates; engineered for high throughput using **Qdrant** and **LangChain** following agile development workflows.
- Developed **FastAPI backend services** with full observability (OpenTelemetry, SigNoz), **unit-tested** and **containerised via Docker**, enabling reliable CI/CD deployment of ML pipelines to production.
- **Tools:** *Python, FastAPI, LangChain, LLMs/SLMs, Qdrant, ClickHouse, Docker, OpenTelemetry*

Accenture — Bengaluru
AEH Intern (Advanced Application Engineering Analyst) [Certificate] — *May 2025 – July 2025*

- Built a **scalable, distributed cloud capacity monitoring system** for a Fortune 100-scale enterprise, processing data across **thousands of Azure subscriptions** with automated alerting and ML-based forecasting.
- Integrated **ML time-series models** into **Azure DevOps CI/CD pipelines** via Azure Functions and Automation, delivering real-time operational insights through **Power BI dashboards**.
- Engineered **end-to-end data pipelines** on **Azure SQL** with relational schema design and rigorous data controls, supporting operational decision-making at scale for a global insurance enterprise.
- **Tools:** *Python, Pandas, Scikit-learn, Azure Functions, Azure Automation, PowerShell, Power BI, Azure SQL, CI/CD*

KGVK Diagnostics — Chennai, India
AI/ML Intern [Certificate] — *Sept 2025 – Nov 2025*

- Developed and deployed a **CNN-based disease classification model** using transfer learning; integrated **Explainable AI** (SHAP/LIME) to ensure model transparency and support clinical decision-making.
- Curated and preprocessed large-scale **facial image datasets**; applied computer vision pipelines end-to-end from data ingestion to production-facing inference using **Python** and **OpenCV**.

Projects

OPTiML-Inspired Self-Supervised Chest X-Ray Analysis Pipeline — *B.Tech Final Year Project*

- Designed and built an **end-to-end ML system** on NIH ChestX-ray14 (100K+ images) spanning SSL pretraining, supervised multi-label classification, localisation, segmentation, and report generation — demonstrating **full software lifecycle ownership** from data pipeline to production deployment.
- Achieved **mean AUC-ROC of 0.8636** across 13 disease classes, outperforming all baselines including ResNet-50 (0.60) and DenseNet-121 (0.77); **per-class threshold optimisation** improved Macro F1 from 0.34 to 0.51, and **50% label efficiency** matched full-data supervised performance.
- Deployed via **FastAPI backend** with modular REST endpoints; pipeline includes disease **bounding box localisation** (Cardiomegaly IoU: 0.66) and **Edge-Aware U-Net segmentation** (Dice: 0.96) feeding a structured radiology report generator.
- **Libraries:** *Python, PyTorch, FastAPI, Sinkhorn OT, LangChain, OpenCV, Pandas, Scikit-learn, Docker*

Achievements

- LeetCode **300+ problems solved** (BiWeekly 183 Rank **2737**); CodeChef **Global Rank 169** in Starters 200 — demonstrating strong **DSA, algorithms, and problem-solving** skills.
- **103rd Rank Nationwide** in Amazon ML Challenge 2025 (Institute Rank 1); **21st Rank Nationwide** Naukri Campus Ring of Honor 2025 (among IIT, NIT, IIIT, BITS students).
- **Responsibilities:** Placement Coordinator, Teaching Assistant – Machine Learning